

Отчёт о лабораторной работе

Лабораторная работа 11

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1 Цель работы

Приобретение практических навыков по настройке удалённого доступа к серверу с помощью SSH.

2 Выполнение лабораторной работы

Для начала зайдём на сервер и авторизуемся от рута, после чего сменим для него пароль (рис. 2.1).

```
[nsandryushin@server.nsandryushin.net ~]$ sudo -i
[sudo] password for nsandryushin:
[root@server.nsandryushin.net ~]# passwd root
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
[root@server.nsandryushin.net ~]#
```

Рис. 2.1: Смена пароля для root

Во втором терминале запустим вывод логов journalctl в реальном времени (рис. 2.2).

```

[nsandryushin@server.nsandryushin.net ~]$ sudo -i
[sudo] password for nsandryushin:
[root@server.nsandryushin.net ~]# journalctl -x -f
Nov 15 16:27:32 server.nsandryushin.net systemd[1]: Starting systemd-hostnamed.service - Hostname Service
...
Subject: A start job for unit systemd-hostnamed.service has begun execution
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

A start job for unit systemd-hostnamed.service has begun execution.

The job identifier is 6935.
Nov 15 16:27:32 server.nsandryushin.net systemd[1]: Started systemd-hostnamed.service - Hostname Service.
Subject: A start job for unit systemd-hostnamed.service has finished successfully
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

A start job for unit systemd-hostnamed.service has finished successfully.

The job identifier is 6935.
Nov 15 16:27:45 server.nsandryushin.net passwd[9307]: pam_unix(passwd:chauthtok): password changed for ro
ot
Nov 15 16:27:45 server.nsandryushin.net passwd[9307]: gkr-pam: couldn't update the login keyring password
: no old password was entered
Nov 15 16:27:53 server.nsandryushin.net systemd[7946]: Started run-p9318-i9618.scope - [systemd-run] /usr
/bin/bash.
Subject: A start job for unit UNIT has finished successfully
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

A start job for unit UNIT has finished successfully.

The job identifier is 611.

```

Рис. 2.2: journalctl

Далее, зайдём на клиент. С клиента попытаемся по ssh подключиться к серверу от имени учётной записи root. Операция успешна (рис. 2.3).

```

[nsandryushin@client.nsandryushin.net ~]$ ssh root@server.nsandryushin.net
root@server.nsandryushin.net's password:
Web console: https://server.nsandryushin.net:9090/ or https://10.0.2.15:9090/

Last failed login: Sat Nov 15 16:28:57 UTC 2025 from 192.168.1.30 on ssh:notty
There was 1 failed login attempt since the last successful login.

```

Рис. 2.3: Авторизация под root по ssh

Посмотрим логи и увидим, что авторизация была разрешена, и пароль был принят (рис. 2.4).

```

Nov 15 16:30:47 server.nsandryushin.net sshd-session[9464]: Accepted password for root from 192.168.1.30
port 32768 ssh2
Nov 15 16:30:47 server.nsandryushin.net systemd-logind[934]: New session 7 of user root.
    Subject: A new session 7 has been created for user root
    Defined-By: systemd
    Support: https://wiki.rockylinux.org/rocky/support
    Documentation: sd-login(3)

    A new session with the ID 7 has been created for the user root.

    The leading process of the session is 9464.
Nov 15 16:30:47 server.nsandryushin.net systemd[1]: Started session-7.scope - Session 7 of User root.
    Subject: A start job for unit session-7.scope has finished successfully
    Defined-By: systemd
    Support: https://wiki.rockylinux.org/rocky/support

    A start job for unit session-7.scope has finished successfully.

    The job identifier is 7684.
Nov 15 16:30:47 server.nsandryushin.net sshd-session[9464]: pam_unix(sshd:session): session opened for us
er root(uid=0) by root(uid=0)
Nov 15 16:30:47 server.nsandryushin.net systemd[1]: Starting systemd-hostnamed.service - Hostname Service
...
    Subject: A start job for unit systemd-hostnamed.service has begun execution
    Defined-By: systemd
    Support: https://wiki.rockylinux.org/rocky/support

    A start job for unit systemd-hostnamed.service has begun execution.

    The job identifier is 7834.
Nov 15 16:30:47 server.nsandryushin.net systemd[1]: Started systemd-hostnamed.service - Hostname Service.
    Subject: A start job for unit systemd-hostnamed.service has finished successfully
    Defined-By: systemd
    Support: https://wiki.rockylinux.org/rocky/support

    A start job for unit systemd-hostnamed.service has finished successfully.

    The job identifier is 7834.

```

Рис. 2.4: Лог авторизации

Теперь откроем на сервере файл `/etc/ssh/sshd_config` и пропишем там `PermitRootLogin` как `no`, для запрета авторизации от имени рута (рис. 2.5).


```
GNU nano 8.1 /etc/ssh/sshd_config
# $OpenBSD: sshd_config,v 1.104 2021/07/02 05:11:21 dtucker Exp $

# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/usr/local/sbin:/usr/sbin

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

# To modify the system-wide sshd configuration, create a *.conf file under
# /etc/ssh/sshd_config.d/ which will be automatically included below
Include /etc/ssh/sshd_config.d/*.conf

# If you want to change the port on a SELinux system, you have to tell
# SELinux about this change.
# semanage port -a -t ssh_port_t -p tcp #PORTNUMBER
#
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10
```

Рис. 2.5: Редактирование /etc/ssh/sshd_config

Перезапустим теперь службу sshd на сервере (рис. 2.6).

```
| [nsandryushin@server.nsandryushin.net ~]$ sudo systemctl restart sshd
```

Рис. 2.6: Перезапуск sshd

Теперь на клиенте попробуем подключиться по ssh под пользователем root и увидим, что пароль неверный. Это не так, но это значит, что авторизация была запрещена (рис. 2.7).

```
[nsandryushin@client.nsandryushin.net ~]$ ssh root@server
The authenticity of host 'server (192.168.1.1)' can't be established.
ED25519 key fingerprint is SHA256:1PtjeoTNJyZ4YiNPkJxzegCAqkN7GCRWznJJBakANJY.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [server.nsandryushin.net]:2022
  ~/.ssh/known_hosts:4: server.nsandryushin.net
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server' (ED25519) to the list of known hosts.
root@server's password:
Permission denied, please try again.
root@server's password:
```

Рис. 2.7: Повторная авторизация под root по ssh

В логах увидим, что авторизация провалена, и не удалось проверить пароль (рис. 2.8).

```
Nov 15 16:33:14 server.nsandryushin.net unix_chkpwd[9551]: password check failed for user (root)
Nov 15 16:33:14 server.nsandryushin.net sshd-session[9549]: pam_unix(sshd:auth): authentication failure;
logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.1.30 user=root
Nov 15 16:33:16 server.nsandryushin.net sshd-session[9549]: Failed password for root from 192.168.1.30 po
rt 42754 ssh2
Nov 15 16:33:18 server.nsandryushin.net sshd-session[9549]: Connection closed by authenticating user root
192.168.1.30 port 42754 [preauth]
```

Рис. 2.8: Лог авторизации

Попробуем подключиться по ssh к серверу от имени своей учётной записи. Как видим, авторизация успешна (рис. 2.9).

```
[nsandryushin@client.nsandryushin.net ~]$ ssh nsandryushin@server.nsandryushin.net
nsandryushin@server.nsandryushin.net's password:
Web console: https://server.nsandryushin.net:9090/ or https://10.0.2.15:9090/

Last login: Sat Nov 15 16:21:42 2025
[nsandryushin@server.nsandryushin.net ~]$ exit
logout
Connection to server.nsandryushin.net closed.
```

Рис. 2.9: Авторизация под своей учётной записью по ssh

Теперь в файле конфигурации sshd пропишем список разрешённых пользова-
телей для авторизации. Пока это будет 1 пользователь - vagrant (рис. 2.10).

```

GNU nano 8.1 /etc/ssh/sshd_config Modified
# $OpenBSD: sshd_config,v 1.104 2021/07/02 05:11:21 dtucker Exp $

# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/usr/local/sbin:/usr/sbin

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

# To modify the system-wide sshd configuration, create a *.conf file under
# /etc/ssh/sshd_config.d/ which will be automatically included below
Include /etc/ssh/sshd_config.d/*.conf

# If you want to change the port on a SELinux system, you have to tell
# SELinux about this change.
# semanage port -a -t ssh_port_t -p tcp #PORTNUMBER
#
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
AllowUsers vagrant

```

Рис. 2.10: Добавление белого списка пользователей

Перезапустим sshd (рис. 2.11).

```

[nsandryushin@server.nsandryushin.net ~]$ sudo systemctl restart sshd

```

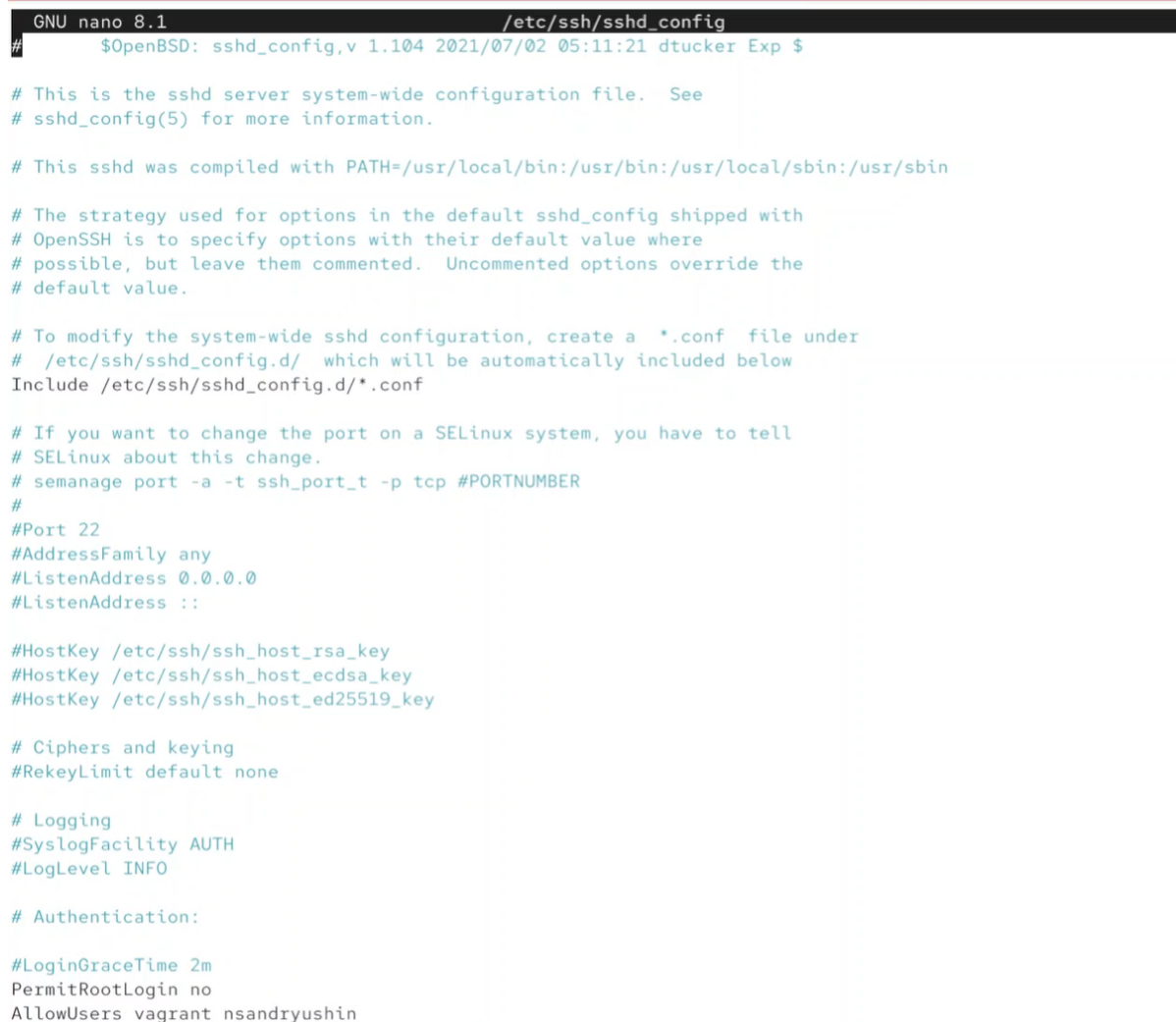
Рис. 2.11: Перезапуск sshd

Вновь на клиенте попробуем авторизоваться под пользователем. Но теперь операция авторизации была провалена, поскольку пользователя нет в белом списке, обозначенном на прошлом этапе (рис. 2.12).

```
[nsandryushin@client.nsandryushin.net ~]$ ssh nsandryushin@server.nsandryushin.net
nsandryushin@server.nsandryushin.net's password:
Permission denied, please try again.
nsandryushin@server.nsandryushin.net's password:
```

Рис. 2.12: Провальная авторизация под своей учётной записью

Снова откроем файл конфигурации на сервере и добавим пользователя в белый список (рис. 2.13).



```
GNU nano 8.1 /etc/ssh/sshd_config
# $OpenBSD: sshd_config,v 1.104 2021/07/02 05:11:21 dtucker Exp $

# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/usr/local/sbin:/usr/sbin

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

# To modify the system-wide sshd configuration, create a *.conf file under
# /etc/ssh/sshd_config.d/ which will be automatically included below
Include /etc/ssh/sshd_config.d/*.conf

# If you want to change the port on a SELinux system, you have to tell
# SELinux about this change.
# semanage port -a -t ssh_port_t -p tcp #PORTNUMBER
#
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
AllowUsers vagrant nsandryushin
```

Рис. 2.13: Добавление нового пользователя в белый

После перезапуска службы мы увидим, что подключение с клиента на сервер под учётной записью пользователя проходит успешно (рис. 2.14).

```
[nsandryushin@client.nsandryushin.net ~]$ ssh nsandryushin@server.nsandryushin.net
nsandryushin@server.nsandryushin.net's password:
Web console: https://server.nsandryushin.net:9090/ or https://10.0.2.15:9090/

Last failed login: Sat Nov 15 16:36:50 UTC 2025 from 192.168.1.30 on ssh:notty
There was 1 failed login attempt since the last successful login.
Last login: Sat Nov 15 16:34:26 2025 from 192.168.1.30
[nsandryushin@server.nsandryushin.net ~]$ exit
logout
Connection to server.nsandryushin.net closed.
```

Рис. 2.14: Успешная авторизация под своей учётной записью

Теперь в файле конфигурации на сервере добавим рабочие порты 22 и 2022 (рис. 2.15).

```
GNU nano 8.1 /etc/ssh/sshd_config Modified
# $OpenBSD: sshd_config,v 1.104 2021/07/02 05:11:21 dtucker Exp $

# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/usr/local/sbin:/usr/sbin

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

# To modify the system-wide sshd configuration, create a *.conf file under
# /etc/ssh/sshd_config.d/ which will be automatically included below
Include /etc/ssh/sshd_config.d/*.conf

# If you want to change the port on a SELinux system, you have to tell
# SELinux about this change.
# semanage port -a -t ssh_port_t -p tcp #PORTNUMBER
#
#Port 22
Port 22
Port 2022
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
AllowUsers vagrant nsandryushin
```

Рис. 2.15: Добавление портов

Перезапустим службу sshd. Выводя статус службы можем увидеть, что служба не смогла привязаться к порту 2022 по причине нехватки прав доступа (рис. 2.16).

```
[nsandryushin@server.nsandryushin.net ~]$ sudo systemctl restart sshd
[nsandryushin@server.nsandryushin.net ~]$ sudo systemctl status -l sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-11-15 16:39:44 UTC; 35s ago
     Invocation: d2dda5e094544f079f26030d9272d8f5
       Docs: man:sshd(8)
             man:sshd_config(5)
    Main PID: 9759 (sshd)
      Tasks: 1 (limit: 10407)
     Memory: 1M (peak: 1.2M)
        CPU: 26ms
    CGroup: /system.slice/sshd.service
            └─9759 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Nov 15 16:39:44 server.nsandryushin.net systemd[1]: Starting sshd.service - OpenSSH server daemon...
Nov 15 16:39:44 server.nsandryushin.net (sshd)[9759]: sshd.service: Referenced but unset environment var>
Nov 15 16:39:44 server.nsandryushin.net sshd[9759]: error: Bind to port 2022 on 0.0.0.0 failed: Permissi>
Nov 15 16:39:44 server.nsandryushin.net sshd[9759]: error: Bind to port 2022 on :: failed: Permission de>
Nov 15 16:39:44 server.nsandryushin.net sshd[9759]: Server listening on 0.0.0.0 port 22.
Nov 15 16:39:44 server.nsandryushin.net systemd[1]: Started sshd.service - OpenSSH server daemon.
Nov 15 16:39:44 server.nsandryushin.net sshd[9759]: Server listening on :: port 22.
lines 1-20/20 (END)
```

Рис. 2.16: Статус службы sshd

Открыв вкладку с работающим journalctl мы можем увидеть, что проблема кроется в SELinux, который не даёт sshd использовать 2022 порт (рис. 2.17).

```

Nov 15 16:39:50 server.nsandryushin.net setroubleshoot[9760]: SELinux is preventing /usr/sbin/sshd from name_bind access on the tcp_socket port 2022. For complete SELinux messages run: sealert -l 341b9b64-7e26-4de2-bc1e-576acf184fc6
Nov 15 16:39:50 server.nsandryushin.net setroubleshoot[9760]: SELinux is preventing /usr/sbin/sshd from name_bind access on the tcp_socket port 2022.

suggests      *****

***** Plugin bind_ports (92.2 confidence)

If you want to allow /usr/sbin/sshd to bind
to network port 2022
Then you need to modify the port type.
Do
# semanage port -a -t PORT_TYPE -p tcp 2022
  where PORT_TYPE is one of the following
: ssh_port_t, vnc_port_t, xserver_port_t.

***** Plugin catchall_boolean (7.83 confidence) suggests *****

If you want to allow nis to enabled
Then you must tell SELinux about this by enabling the 'nis_enabled' boolean.

Do
setsebool -P nis_enabled 1

***** Plugin catchall (1.41 confidence) suggests *****

If you believe that sshd should be allowed
name_bind access on the port 2022 tcp_socket by default.
Then you should report this as a bug.
You can generate a local policy module to allow this access.

Do
allow this access for now by executing:
# ausearch -c 'sshd' --raw | audit2allow -M my-sshd
# semodule -X 300 -i my-sshd.pp

Nov 15 16:39:53 server.nsandryushin.net setroubleshoot[9760]: SELinux is preventing /usr/sbin/sshd from name_bind access on the tcp_socket port 2022. For complete SELinux messages run: sealert -l 341b9b64-7e26-4de2-bc1e-576acf184fc6
Nov 15 16:39:53 server.nsandryushin.net setroubleshoot[9760]: SELinux is preventing /usr/sbin/sshd from name_bind access on the tcp_socket port 2022.

```

Рис. 2.17: Лог с SELinux

Настроим SELinux метки так, чтобы разрешить sshd работать с 2022 портом и также настроим фаервол, разрешив пропускать трафик по 2022 порту, после чего перезапустим sshd и выведем его статус. Как видим, служба теперь прослушивает 2 порта - 22 и 2022 (рис. 2.18).


```

[nsandryushin@server.nsandryushin.net ~]$ sudo semanage port -a -t ssh_port_t -p tcp 2022
[nsandryushin@server.nsandryushin.net ~]$ sudo firewall-cmd --add-port=2022/tcp
success
[nsandryushin@server.nsandryushin.net ~]$ firewall-cmd --add-port=2022/tcp --permanent
success
[nsandryushin@server.nsandryushin.net ~]$ sudo systemctl restart sshd
[nsandryushin@server.nsandryushin.net ~]$ sudo systemctl status -l sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-11-15 16:42:23 UTC; 10s ago
 Invocation: 6d8c72f73fe043129be5ef486b214365
    Docs: man:sshd(8)
          man:sshd_config(5)
 Main PID: 9897 (sshd)
   Tasks: 1 (limit: 10407)
  Memory: 1M (peak: 1.3M)
     CPU: 25ms
  CGroup: /system.slice/sshd.service
          └─9897 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Nov 15 16:42:23 server.nsandryushin.net systemd[1]: Starting sshd.service - OpenSSH server daemon...
Nov 15 16:42:23 server.nsandryushin.net (sshd)[9897]: sshd.service: Referenced but unset environment var>
Nov 15 16:42:23 server.nsandryushin.net sshd[9897]: Server listening on 0.0.0.0 port 2022.
Nov 15 16:42:23 server.nsandryushin.net sshd[9897]: Server listening on :: port 2022.
Nov 15 16:42:23 server.nsandryushin.net systemd[1]: Started sshd.service - OpenSSH server daemon.
Nov 15 16:42:23 server.nsandryushin.net sshd[9897]: Server listening on 0.0.0.0 port 22.
Nov 15 16:42:23 server.nsandryushin.net sshd[9897]: Server listening on :: port 22.

```

Рис. 2.18: Настройка SELinux и firewall

Теперь с клиента попробуем подключиться к серверу по ssh, используя добавленные нами порты. Как видим, оба подключения прошли успешно (рис. 2.19).

```

[nsandryushin@client.nsandryushin.net ~]$ ssh nsandryushin@server.nsandryushin.net
nsandryushin@server.nsandryushin.net's password:
Web console: https://server.nsandryushin.net:9090/ or https://10.0.2.15:9090/

Last login: Sat Nov 15 16:38:12 2025 from 192.168.1.30
[nsandryushin@server.nsandryushin.net ~]$ sudo -i
[sudo] password for nsandryushin:
[root@server.nsandryushin.net ~]# logout
[nsandryushin@server.nsandryushin.net ~]$ logout
Connection to server.nsandryushin.net closed.
[nsandryushin@client.nsandryushin.net ~]$ ssh -p2022 nsandryushin@server.nsandryushin.net
nsandryushin@server.nsandryushin.net's password:
Web console: https://server.nsandryushin.net:9090/ or https://10.0.2.15:9090/

Last login: Sat Nov 15 16:43:15 2025 from 192.168.1.30
[nsandryushin@server.nsandryushin.net ~]$ sudo -i
[sudo] password for nsandryushin:
[root@server.nsandryushin.net ~]# logout
[nsandryushin@server.nsandryushin.net ~]$ logout
Connection to server.nsandryushin.net closed.

```

Рис. 2.19: Проверка портов

Вновь откроем файл конфигурации и разрешим авторизацию по ключу (рис. 2.20).


```

GNU nano 8.1 /etc/ssh/sshd_config
# $OpenBSD: sshd_config,v 1.104 2021/07/02 05:11:21 dtucker Exp $

# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/usr/local/sbin:/usr/sbin

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

# To modify the system-wide sshd configuration, create a *.conf file under
# /etc/ssh/sshd_config.d/ which will be automatically included below
Include /etc/ssh/sshd_config.d/*.conf

# If you want to change the port on a SELinux system, you have to tell
# SELinux about this change.
# semanage port -a -t ssh_port_t -p tcp #PORTNUMBER
#
#Port 22
Port 22
Port 2022
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
AllowUsers vagrant nsandryushin
PubkeyAuthentication yes
#StrictModes yes

```

Рис. 2.20: Включение авторизации по ключу

Перезапустим sshd (рис. 2.21).

```

[nsandryushin@server.nsandryushin.net ~]$ sudo systemctl restart sshd

```

Рис. 2.21: Перезапуск sshd

Теперь на клиенте сгенерируем rsa ключ и отправим его на сервер (рис. 2.22).

```
[nsandryushin@client.nsandryushin.net ~]$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/nsandryushin/.ssh/id_ed25519): ^C
[nsandryushin@client.nsandryushin.net ~]$ ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/home/nsandryushin/.ssh/id_rsa):
Enter passphrase for "/home/nsandryushin/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/nsandryushin/.ssh/id_rsa
Your public key has been saved in /home/nsandryushin/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:ZfBX3Z0Km9unHkxg3r65I4RtNc7uBxLNQYmwmKbWGcs nsandryushin@client.nsandryushin.net
The key's randomart image is:
+---[RSA 3072]-----+
|      ... o.o.+      |
|      oo.o + .+      |
|      = .+oB o       |
|      = +oo=o*       |
|      o ES o.Bo.     |
|      . . *+= .      |
|      o o++         |
|      . ++.         |
|      ++           |
+-----[SHA256]-----+
[nsandryushin@client.nsandryushin.net ~]$ ssh-copy-id nsandryushin@server.nsandryushin.net
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ssh-add -L
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already
installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install t
he new keys
nsandryushin@server.nsandryushin.net's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'nsandryushin@server.nsandryushin.net'"
and check to make sure that only the key(s) you wanted were added.
```

Рис. 2.22: Формирование ключа для авторизации

Теперь попробуем подключиться к серверу по ssh, и на этот раз у нас не спросят пароль (рис. 2.23).

```

[nsandryushin@client.nsandryushin.net ~]$ ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/home/nsandryushin/.ssh/id_rsa):
Enter passphrase for "/home/nsandryushin/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/nsandryushin/.ssh/id_rsa
Your public key has been saved in /home/nsandryushin/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:ZfBX3Z0Km9unHkxg3r65I4RtNc7uBxLNQYmwmKbWGcs nsandryushin@client.nsandryushin.net
The key's randomart image is:
+---[RSA 3072]-----+
|      . . . o.o.+ |
|      oo.o + .+ |
|      = .+oB o |
|      = +oo=o* |
|      o ES o.Bo. |
|      . . *+= . |
|      o o++ |
|      . ++. |
|      +*+ |
+-----[SHA256]-----+
[nsandryushin@client.nsandryushin.net ~]$ ssh-copy-id nsandryushin@server.nsandryushin.net
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ssh-add -L
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already
installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install t
he new keys
nsandryushin@server.nsandryushin.net's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'nsandryushin@server.nsandryushin.net'"
and check to make sure that only the key(s) you wanted were added.

[nsandryushin@client.nsandryushin.net ~]$ ssh nsandryushin@server.nsandryushin.net
Web console: https://server.nsandryushin.net:9090/ or https://10.0.2.15:9090/

Last login: Sat Nov 15 16:44:03 2025 from 192.168.1.30
[nsandryushin@server.nsandryushin.net ~]$
logout
Connection to server.nsandryushin.net closed.

```

Рис. 2.23: Подключение без пароля

Теперь выведем список служб, использующих TCP (рис. 2.24).

```
[nsandryushin@client.nsandryushin.net ~]$ lsof | grep TCP
firefox      8183                nsandryushin    73u    IPv4                38130      0t0      TCP client
.nsandryushin.net:46436->93.243.107.34.bc.googleusercontent.com:https (ESTABLISHED)
firefox      8183 8203 firefox      nsandryushin    73u    IPv4                38130      0t0      TCP client
.nsandryushin.net:46436->93.243.107.34.bc.googleusercontent.com:https (ESTABLISHED)
firefox      8183 8204 WaylandPr    nsandryushin    73u    IPv4                38130      0t0      TCP client
.nsandryushin.net:46436->93.243.107.34.bc.googleusercontent.com:https (ESTABLISHED)
firefox      8183 8205 pool-spaw     nsandryushin    73u    IPv4                38130      0t0      TCP client
.nsandryushin.net:46436->93.243.107.34.bc.googleusercontent.com:https (ESTABLISHED)
firefox      8183 8206 gmain         nsandryushin    73u    IPv4                38130      0t0      TCP client
.nsandryushin.net:46436->93.243.107.34.bc.googleusercontent.com:https (ESTABLISHED)
firefox      8183 8208 gdbus         nsandryushin    73u    IPv4                38130      0t0      TCP client
.nsandryushin.net:46436->93.243.107.34.bc.googleusercontent.com:https (ESTABLISHED)
firefox      8183 8209 dconf\x20       nsandryushin    73u    IPv4                38130      0t0      TCP client
.nsandryushin.net:46436->93.243.107.34.bc.googleusercontent.com:https (ESTABLISHED)
firefox      8183 8210 glean.dis      nsandryushin    73u    IPv4                38130      0t0      TCP client
.nsandryushin.net:46436->93.243.107.34.bc.googleusercontent.com:https (ESTABLISHED)
firefox      8183 8212 IPC\x20I/     nsandryushin    73u    IPv4                38130      0t0      TCP client
.nsandryushin.net:46436->93.243.107.34.bc.googleusercontent.com:https (ESTABLISHED)
firefox      8183 8213 Timer         nsandryushin    73u    IPv4                38130      0t0      TCP client
.nsandryushin.net:46436->93.243.107.34.bc.googleusercontent.com:https (ESTABLISHED)
firefox      8183 8214 Netlink       nsandryushin    73u    IPv4                38130      0t0      TCP client
.nsandryushin.net:46436->93.243.107.34.bc.googleusercontent.com:https (ESTABLISHED)
firefox      8183 8215 Socket         nsandryushin    73u    IPv4                38130      0t0      TCP client
.nsandryushin.net:46436->93.243.107.34.bc.googleusercontent.com:https (ESTABLISHED)
firefox      8183 8221 IPDL\x20B      nsandryushin    73u    IPv4                38130      0t0      TCP client
.nsandryushin.net:46436->93.243.107.34.bc.googleusercontent.com:https (ESTABLISHED)
```

Рис. 2.24: Список служб, использующих TCP

Теперь перенаправим порт 8080 на server.nsandryushin.net на 80 порт локалхоста (рис. 2.25).

```
[nsandryushin@client.nsandryushin.net ~]$ ssh -fNL 8080:localhost:80 nsandryushin@server.nsandryushin.net
[nsandryushin@client.nsandryushin.net ~]$ lsof | grep TCP
firefox      8183                nsandryushin    73u    IPv4                38130      0t0      TCP client
.nsandryushin.net:46436->93.243.107.34.bc.googleusercontent.com:https (ESTABLISHED)
firefox      8183 8203 firefox      nsandryushin    73u    IPv4                38130      0t0      TCP client
.nsandryushin.net:46436->93.243.107.34.bc.googleusercontent.com:https (ESTABLISHED)
firefox      8183 8204 WaylandPr    nsandryushin    73u    IPv4                38130      0t0      TCP client
.nsandryushin.net:46436->93.243.107.34.bc.googleusercontent.com:https (ESTABLISHED)
firefox      8183 8205 pool-spaw     nsandryushin    73u    IPv4                38130      0t0      TCP client
.nsandryushin.net:46436->93.243.107.34.bc.googleusercontent.com:https (ESTABLISHED)
```

Рис. 2.25: Перенаправление порта

В списке TCP служб мы увидим ssh, который перенаправляет трафик (рис. 2.26).

```
firefox      8183 9130 StreamTra    nsandryushin    73u    IPv4                38130      0t0      TCP client
.nsandryushin.net:46436->93.243.107.34.bc.googleusercontent.com:https (ESTABLISHED)
firefox      8183 9131 StreamTra    nsandryushin    73u    IPv4                38130      0t0      TCP client
.nsandryushin.net:46436->93.243.107.34.bc.googleusercontent.com:https (ESTABLISHED)
ssh          9136                nsandryushin     3u    IPv4                46246      0t0      TCP client
.nsandryushin.net:42016->mail.nsandryushin.net:ssh (ESTABLISHED)
ssh          9136                nsandryushin     4u    IPv6                46255      0t0      TCP localh
ost:webcache (LISTEN)
ssh          9136                nsandryushin     5u    IPv4                46256      0t0      TCP localh
ost:webcache (LISTEN)
```

Рис. 2.26: Список служб

Теперь, обращаясь к localhost:8080 мы видим наш сайт по адресу server.nsandryushin.net:80 (рис. 2.27).

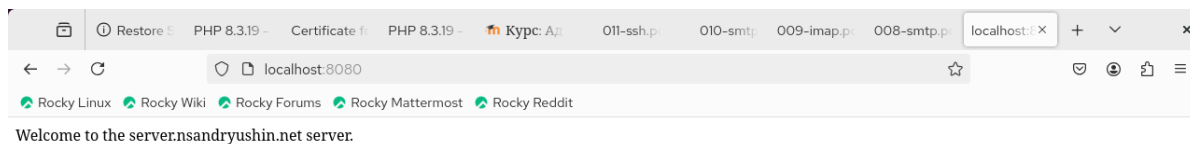


Рис. 2.27: Результат переадресации

На клиенте теперь попробуем вызвать несколько консольных утилит через ssh на сервере, а именно вывод имени хоста, вывод содержимого домашнего каталога и вывод почты (рис. 2.28).

```
[nsandryushin@client.nsandryushin.net ~]$ ssh nsandryushin@server.nsandryushin.net hostname
server.nsandryushin.net
[nsandryushin@client.nsandryushin.net ~]$ ssh nsandryushin@server.nsandryushin.net ls -Al
total 32
-rw-----. 1 nsandryushin nsandryushin 418 Nov 15 16:44 .bash_history
-rw-r--r--. 1 nsandryushin nsandryushin 18 Oct 29 2024 .bash_logout
-rw-r--r--. 1 nsandryushin nsandryushin 144 Oct 29 2024 .bash_profile
-rw-r--r--. 1 nsandryushin nsandryushin 549 Sep 6 18:17 .bashrc
drwx-----. 12 nsandryushin nsandryushin 4096 Oct 25 16:16 .cache
drwx-----. 12 nsandryushin nsandryushin 4096 Oct 25 16:16 .config
drwxr-xr-x. 2 nsandryushin nsandryushin 6 Sep 13 13:01 Desktop
drwxr-xr-x. 2 nsandryushin nsandryushin 6 Sep 13 13:01 Documents
drwxr-xr-x. 2 nsandryushin nsandryushin 6 Sep 13 13:01 Downloads
-rw-r-----. 1 root root 3033 Sep 20 12:48 kea-dhcp-ddns.conf
drwx-----. 4 nsandryushin nsandryushin 32 Sep 13 13:01 .local
drwx-----. 5 nsandryushin nsandryushin 4096 Nov 8 00:18 Maildir
drwxr-xr-x. 5 nsandryushin nsandryushin 54 Sep 13 13:02 .mozilla
drwxr-xr-x. 2 nsandryushin nsandryushin 6 Sep 13 13:01 Music
drwxr-xr-x. 2 nsandryushin nsandryushin 6 Sep 13 13:01 Pictures
drwxr-xr-x. 2 nsandryushin nsandryushin 6 Sep 13 13:01 Public
drwx-----. 2 nsandryushin nsandryushin 29 Nov 15 16:47 .ssh
drwxr-xr-x. 2 nsandryushin nsandryushin 6 Sep 13 13:01 Templates
drwxr-xr-x. 2 nsandryushin nsandryushin 6 Sep 13 13:01 Videos
[nsandryushin@client.nsandryushin.net ~]$ ssh nsandryushin@server.nsandryushin.net MAIL=~/.Maildir/ mail
s-nail version v14.9.24. Type '?' for help
/home/nsandryushin/Maildir: 3 messages 2 unread
 1 Nikita Andryushin 2025-10-31 19:49 18/705 "test"
►U 2 nsandryushin@client. 2025-11-07 23:27 21/901 "LMTP test"
U 3 Nikita Andryushin 2025-11-08 00:17 21/885 "lab10test"
```

Рис. 2.28: Запуск консольных утилит по ssh

В конфигурационном файле sshd на сервере включим форвардинг X11 (рис. 2.29).

```
GNU nano 8.1 /etc/ssh/sshd_config Modified

# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/usr/local/sbin:/usr/sbin

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

# To modify the system-wide sshd configuration, create a *.conf file under
# /etc/ssh/sshd_config.d/ which will be automatically included below
Include /etc/ssh/sshd_config.d/*.conf

# If you want to change the port on a SELinux system, you have to tell
# SELinux about this change.
# semanage port -a -t ssh_port_t -p tcp #PORTNUMBER
#
#Port 22
Port 22
Port 2022
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
AllowUsers vagrant nsandryushin
PubkeyAuthentication yes
X11Forwarding yes
#StrictModes yes
```

Рис. 2.29: Включение форвардинга иксов

Перезапустим sshd (рис. 2.30).

```
[nsandryushin@server.nsandryushin.net ~]$ sudo systemctl restart sshd
```

Рис. 2.30: Перезапуск sshd

И запустим графическое приложение, например firefox, по ssh (рис. 2.31).

```
[nsandryushin@client.nsandryushin.net ~]$ ssh -YC nsandryushin@server.nsandryushin.net firefox
/usr/bin/xauth: file /home/nsandryushin/.Xauthority does not exist
[nsandryushin@client.nsandryushin.net ~]$
```

Рис. 2.31: Запуск графического приложения по ssh

Как видим, на сервере запустился firefox (рис. 2.32).

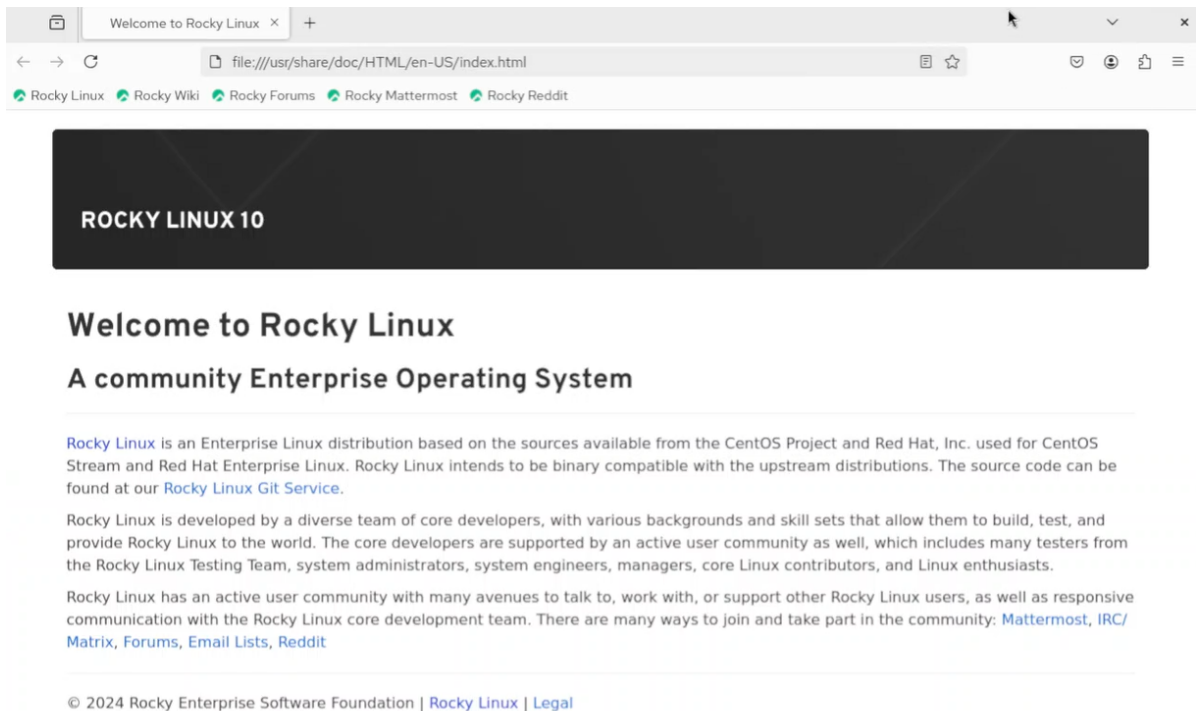


Рис. 2.32: Результат запуска графического приложения по ssh

Теперь сохраним файлы, с которыми велась работа, в vagrant (рис. 2.33).

```
[nsandryushin@server.nsandryushin.net ~]$ cd /vagrant/provision/server
[nsandryushin@server.nsandryushin.net server]$ sudo mkdir -p /vagrant/provision/server/ssh/etc/ssh
[nsandryushin@server.nsandryushin.net server]$ sudo cp -R /etc/ssh/sshd_config /vagrant/provision/server/ssh/etc/ssh/
[nsandryushin@server.nsandryushin.net server]$ cd /vagrant/provision/server
[nsandryushin@server.nsandryushin.net server]$ sudo touch ssh.sh
[nsandryushin@server.nsandryushin.net server]$ sudo chmod +x ssh.sh
[nsandryushin@server.nsandryushin.net server]$ nano ssh.sh
[nsandryushin@server.nsandryushin.net server]$ sudo nano ssh.sh
[nsandryushin@server.nsandryushin.net server]$
```

Рис. 2.33: Сохранение в vagrant

Создадим скрипт `ssh.sh` и наполним его следующим содержанием (рис. 2.34).

```
GNU nano 8.1                                ssh.sh                                Modified
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/ssh/etc/* /etc
restorecon -vR /etc
echo "Configure firewall"
firewall-cmd --add-port=2022/tcp
firewall-cmd --add-port=2022/tcp --permanent
echo "Tuning SELinux"
semanage port -a -t ssh_port_t -p tcp 2022
echo "Restart sshd service"
systemctl restart sshd
```

Рис. 2.34: `ssh.sh`

И добавим этот скрипт в `vagrantfile` (рис. 2.35).

```
97     server.vm.provision "server firewall",
98     type: "shell",
99     preserve_order: true,
100    path: "provision/server/firewall.sh"
101    server.vm.provision "server mail",
102    type: "shell",
103    preserve_order: true,
104    path: "provision/server/mail.sh"
105    server.vm.provision "server ssh",
106    type: "shell",
107    preserve_order: true,
108    path: "provision/server/ssh.sh"
109
110 end
```

Рис. 2.35: `Vagrantfile`

3 Выводы

В результате выполнения лабораторной работы были получены навыки настройки ssh, форвардинга через него и настройки авторизации